



ADITYA RAJ | 25AG65R08
AQUACULTURAL ENGINEERING

EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2027	M.TECH	IIT Kharagpur	8.37 / 10
2025	B. Tech(Agril. Engg.)	Vasantrao Naik Marathwada Krishi Vidyapeeth, Parbhani	7.93 / 10
2019	Senior Secondary Examination	Bihar School Examination Board, Patna	76.4%
2017	Secondary School Examination	Central Board Of Secondary Education	9.2 / 10

PROJECTS

- Built End-to-End Multi-SKU Retail Demand Forecasting & Optimization System (Hierarchical ML + MLOps)** [Self Project]
- Built a scalable **30,000+** SKU hierarchical demand forecasting pipeline using **SQL**, **LightGBM** quantile regression, and engineered lag, rolling & calendar features; improved **WRMSSE** by **~0.97%** over baseline models through Optuna-driven tuning and time-series cross-validation.
 - Built a production-ready forecasting system using **Docker + FastAPI**, integrating **MLflow** for experiment tracking, automated retraining, and model-drift monitoring to support scalable, data-driven inventory and replenishment decisions across complex, high-volume operations.
- Credit Risk Scoring & Cost-Sensitive Decision Optimization System** [Self Project]
- Built a credit risk scoring system on **1.34M+ LendingClub** records using **XGBoost**, **WoE/IV** based feature selection, and temporal validation; improved **PR-AUC** by **~14%** over logistic regression while achieving a **KS statistic** value of **~0.33**, enabling reliable risk discrimination.
 - Optimized cost-sensitive loan approval thresholds using robust, data-driven decisioning & risk analysis, reducing estimated **decision cost** by **~25%** and generating **SHAP-based reason codes** for transparent, interpretable & reliable risk decisions across diverse borrower segments.
- Stock Price Prediction using LSTM (Long Short-Term Memory) based time series modeling and multi-step forecasting** [Self Project]
- Built **LSTM** time-series forecasting model on **4.7K+ S&P 500** observations, using **50-step sequences** of price & volume with an **85:15 train-test split**
 - Engineered **3-layer LSTM** with **100 units/layer** & **20% dropout** implementing multi-step forecasting & evaluating performance using **MSE & RMSE**
- Impact of Soil Quality and Weather Parameters on Crop Growth Analysis using Python & Machine Learning** [Self Project]
- Analyzed **2.2K+** agricultural records across 7 soil & weather parameters using **Python**, identifying crop-specific nutrient and climatic patterns.
 - Executed **EDA & correlation** analysis with **6+ visualization techniques**, deriving nutrient-weather insights for crop suitability & recommendation
- NEXA(Next-Generation Enhanced Execution Assistant): Voice-Activated Virtual Assistant Using Python** [Self Project]
- Developed a **Python-based voice-activated** virtual assistant integrating **speech recognition** technologies for intelligent user interaction
 - Enabled **voice-controlled automation** of routine tasks using speech recognition and command processing for efficient command execution

SKILLS AND EXPERTISE

Software & Tools : SolidWorks | MATLAB | ArcGIS | QGIS | Jupyter Notebook | VS Code | MS Office | Git | GitHub | Power BI | PostgreSQL
Language & Libraries : C | C++ | Python | NumPy | Pandas | Matplotlib | Seaborn | Scikit-learn | TensorFlow | Keras | Django | LightGBM
Core Skills : Machine Learning | Deep Learning | Data Structures & Algorithms | Statistical Analysis | SQL | EDA | Feature Engineering

CERTIFICATIONS

- Supervised Machine Learning: Regression and Classification**
- Programmed and fine-tuned supervised Machine Learning models using regression, classification, regularization, and logistic regression
 - Optimized ML performance through cross-validation, bias-variance analysis, overfitting control & gradient descent using Python & sklearn
- Earned **Deep Learning & TensorFlow** certification, building **neural networks** using backpropagation, activation functions and model evaluation
- Earned **Python for Data Science** certification, engineering data-analysis workflows using **Python, NumPy and Pandas** for preprocessing and EDA
- Leveraged **SQL & Database Management**, writing complex queries using **joins, subqueries, aggregations, CTEs and window functions** for data analysis
- Developed expertise in **Data Visualization & Power BI**, engineering interactive dashboards using **KPIs, slicers, Power Query & advanced visualization**
- AutoCAD Drafting & Design (2D & Basic 3D)** Aditi CAD Training Centre, Parbhani [Sep'24-Oct'24]
- Developed hands-on proficiency in **2D drafting, technical drawings, precise dimensioning, layers, and detailed layout preparation** using AutoCAD.
 - Practiced basic **3D modeling, object creation, editing commands, and drawing modifications** for technical design and engineering applications.

POSITIONS OF RESPONSIBILITY

- Placement Coordinator | Self Specialization | IIT Kharagpur** [Jul'25-Present]
- Coordinated **placement activities**, liaising with recruiters, faculty and students to streamline recruitment processes and communication
 - Spearheaded **industry-student engagement**, managing placement drives, employer interactions, activities and stakeholder coordination
- Teaching Assistant (TA) | Agricultural & Food Engineering, IIT Kharagpur** [Sep'26-Present]
- Supported faculty in **academic delivery**, coordinating laboratory sessions, resolving student queries & managing course-related activities
 - Enhanced **student learning**, coordinating practical sessions, clarifying technical concepts & supporting academic and laboratory activities
- Hall Secretary (Technology) | LBS Hall IIT Kharagpur** [Aug'25-May'26]
- Spearheaded **technical activities** & led cross-functional teams at the hall level, driving **technology initiatives** and coordinating execution

EXTRA CURRICULAR ACTIVITIES

- Smart India Hackathon (SIH), IIT Kharagpur** [Dec'2025]
- Contributed to **organizing** and executing a **national-level hackathon**, coordinating event operations, participant engagement and activities
- NSS Volunteer- Participated in State Level Training Camp on Disaster Management by NSS Wing, KBCNMU Jalgaon, Maharashtra** [Dec'2022]